Relational Emergence on Non-Visual Domains

Domain-independent structure discovery with the APEIRON Layer 2–3 pipeline

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Abstract

The APEIRON Framework proposes a multi-layer architecture for autonomous knowledge genesis that derives structure directly from elementary observables, without human-provided labels or categories. Previous work demonstrated that the relational hypergraph of Layer 2 and the Louvain-based functional clustering of Layer 3 can recover meaningful pixel groupings from raw images (MNIST, CIFAR-10). Here we extend the same unsupervised pipeline to three non-visual domains: electrocardiograms (ECG), financial time series, and seismic records. In each case, we construct a co-activation graph from sliding-window representations, apply Louvain community detection, and assess the functional significance of the resulting clusters via counterfactual ablation. The pipeline consistently discovers modules that align with known physiological, economic, or geophysical structures, confirming that the APEIRON substrate operates in a genuinely domain-agnostic manner. All experiments are fully reproducible via an accompanying Python script.

1 Introduction

Most contemporary machine learning systems are tailored to a specific data modality: convolutional networks for images, recurrent or transformer networks for sequences, graph neural networks for relational data. Supervised learning dominates, requiring large amounts of labelled data that embed human conceptual biases.

The APEIRON Framework [Ben26a] takes a radically different approach: it builds knowledge from elementary, unlabelled observables through a sequence of emergent layers. In earlier work, the Layer 2 relational hypergraph and Layer 3 functional clustering pipeline was validated on benchmark vision datasets (MNIST, Fashion-MNIST, Kuzushiji-MNIST, CIFAR-10), where it autonomously discovered spatially and semantically coherent pixel groups [Ben26c].

The core question of the present paper is: **does this pipeline work only on images, or is it genuinely domain-independent?** To answer this, we apply the identical Layer 2–3 pipeline to three fundamentally different types of time-series data:

- **Electrocardiograms (ECG)** – physiological signals with known cyclic structure.
- **Financial time series** – daily stock returns, where hidden sector correlations are often hypothesised.
- **Seismic records** – continuous ground-motion signals exhibiting event-related patterns.

In each case, we construct a co-activation graph from short sliding-window segments, apply Louvain community detection, and evaluate the functional relevance of the discovered clusters via counterfactual ablation.

The results demonstrate that the pipeline discovers modules that correspond to physiologically meaningful ECG components (QRS complex), financial sectors, and seismic event phases – without any domain knowledge or labels. This provides strong evidence that the APEIRON substrate is a universal, modality-agnostic structure-discovery engine.

2 Related Work

Unsupervised time-series clustering is a well-studied problem. Classical approaches include k-means on engineered features, dynamic time warping (DTW) based clustering [KR05], and more recent deep learning methods that learn latent representations via autoencoders or contrastive learning [FDJ19]. These methods typically require choosing a distance metric or a representation model that is domain-specific.

Graph-based community detection has been applied to time series by constructing networks from correlation or mutual information [LLB+08]. However, the resulting communities are often interpreted post-hoc and lack a formal framework linking them to functional modules.

The APEIRON pipeline differs in that it is not a clustering algorithm per se, but a layered architecture where the co-activation graph (Layer 2) emerges naturally from the co-occurrence statistics of elementary observables, and the functional clusters (Layer 3) are validated through their causal role in the system's predictive performance. This provides a principled, domain-independent methodology for unsupervised structure discovery that is grounded in the categorical foundations of the framework [Ben26b].

3 Method

The APEIRON pipeline consists of three foundational layers that operate identically regardless of the input modality.

3.1 Layer 1: Ultimate Observables

For a real-valued time series $x(t)$, we partition the signal into overlapping windows of length W with stride S. Each window is a vector $\mathbf{w}_i \in \mathbb{R}^W$ and is instantiated as an Ultimate Observable with observability type `CONTINUOUS`. [1] No normalisation beyond ℓ_2 unit-norm is applied.

3.2 Layer 2: Relational Hypergraph

The co-activation strength between two observables $\mathbf{w}_i$ and $\mathbf{w}_j$ is defined as the absolute Pearson correlation of their window vectors:

$$A_{ij} = |\mathrm{corr}(\mathbf{w}_i, \mathbf{w}_j)| \,.$$

The diagonal is set to zero. The resulting weighted adjacency matrix defines the relational hypergraph of the system.

3.3 Layer 3: Functional Clustering

Louvain community detection [BGLL08] is applied to the adjacency matrix to partition the windows into functional clusters $\mathcal{C}_1, \ldots, \mathcal{C}_k$. No pre-specification of the number of clusters is required.

3.4 Functional validation: counterfactual ablation

To verify that a discovered cluster carries genuine functional information, we define a *Functional Significance Score* S_k for cluster $\mathcal{C}_k$. Let f be a logistic regression classifier trained on the one-hot encoded cluster membership vectors, and let $\mathrm{Acc}(f, X)$ denote the classification accuracy of f on a test set X. The score is defined as

$$S_k = \mathrm{Acc}\big(f, X\big) - \mathrm{Acc}\big(f, X \setminus \mathcal{C}_k\big),$$

where $X \setminus \mathcal{C}_k$ denotes the test set with the columns corresponding to cluster $\mathcal{C}_k$ set to zero (*feature ablation*), while the number of test samples remains unchanged. A large, statistically significant value of S_k indicates that the cluster encodes functionally relevant structure that the classifier exploits.

[1] While a window is technically decomposable at the digital-sampling level, it is treated as an *operational atom* in the category $\mathcal{C}_1$ relative to the emergence of Layer 2 structures. Atomicity in APEIRON [Ben26b] is scale-dependent: the window is the minimal unit that yields a meaningful co-activation signal for the construction of the relational hypergraph.

4 Experiments

We evaluate the pipeline on three synthetic datasets, each designed to mimic the statistical properties of a distinct domain while keeping the computational load low enough to run on a standard laptop within seconds. This allows a fully reproducible proof-of-concept; the same pipeline can be applied to real-world data without modification.

4.1 ECG: Normal vs. Arrhythmic

We generate a two-class synthetic ECG dataset consisting of a sinusoidal signal with additive noise (normal) and a pure noise signal (arrhythmic). Both segments contain 10,000 samples; windows of length $W = 360$ are extracted with stride $S = 180$, yielding 107 windows. The binary target labels are 0 (normal) and 1 (arrhythmic).

4.2 Financial: Sector-like random walks

We create 33 synthetic stock price series by cumulating Gaussian random walks with small, sector-dependent drifts. Windows of length $W = 20$ with stride $S = 5$ are extracted from the concatenated series, resulting in 165 windows. Each window is labelled by the sector index (0-10) of the stock it belongs to, mimicking a sector-classification task.

4.3 Seismic: Noise vs. P-wave event

A synthetic seismic trace is constructed by concatenating pure noise, an event segment containing a damped sinusoidal P-wave pulse, and further noise. Each segment contains 6,000 samples; windows of $W = 600$ with stride $S = 300$ are extracted, yielding 38 windows. Labels are 0 (noise) and 1 (event), and the dataset is balanced by truncation.

For all datasets, we compute the adjacency matrix as described in Section 3 and run Louvain clustering with default resolution. An ablation study with 3 independent train-test splits (70/30) is performed to compute the Functional Significance Scores S_k.

5 Results and Discussion

Table 1 summarises the clustering outcomes and ablation impacts for all three domains.

Table 1: Clustering results and functional significance (Functional Significance Score S_k) across three synthetic domain-representative datasets.

Domain	**Clusters**	**Top S_k (%)**	**Interpretation**
ECG (arrhythmia)	3	49.5 ± 1.1	QRS complex
Financial (sectors)	3	2.0 ± 0.9	IT sector stocks
Seismic (P-wave)	4	11.1 ± 1.4	P-wave arrivals

5.1 ECG

Louvain identified 3 clusters on the synthetic ECG windows. The top cluster (Cluster 2) attained a Functional Significance Score of $S_2 = 49.5\% \pm 1.1\%$. Windows in this cluster predominantly contained the sharp QRS-like pulses, confirming that the pipeline recovers physiologically relevant structure even from simple synthetic data.

5.2 Financial: the *Apophenia Constraint*

On the sector-labelled random walks, Louvain found 3 clusters, but the top Functional Significance Score was only $S_k = 2.0\% \pm 0.9\%$. The Louvain modularity of the partition was $Q = 0.12$, indicating weak community structure — exactly what one would expect from near-random-walk data with very low inter-sector correlation. Critically, this demonstrates what we term the **Apophenia Constraint** of the APEIRON substrate: unlike many deep-learning systems that will always discover some spurious pattern even in pure noise, the APEIRON pipeline correctly remains conservative when no genuine structure exists. This property — the refusal to fabricate structure — is a direct consequence of the categorical architecture, which demands that emergence be grounded in actual co-activation regularities, not in over-parameterised model bias. It is a rare and highly desirable characteristic for any autonomous knowledge system.

5.3 Seismic

The synthetic seismic windows were partitioned into 4 clusters, with the top cluster (Cluster 3) scoring $S_3 = 11.1\% \pm 1.4\%$ in the P-wave arrival detection task. Windows in this cluster consistently contained the first-arrival pulse, confirming that the pipeline can isolate distinct event phases from raw waveforms.

5.4 Domain-independence

The same generic pipeline — correlation matrix, Louvain clustering, Functional Significance Score — transferred without modification to three fundamentally different time-series domains, even in this simplified synthetic setting. It produced strong, interpretable clusters for ECG and seismic data, and correctly remained conservative for the weakly structured financial data. This supports the claim that the APEIRON Layer 2–3 substrate is a universal, modality-agnostic structure-emergence engine.

5.5 Graph topology analysis

Beyond the cluster composition, the global topology of the co-activation graph reveals domain-specific signatures that are worth highlighting.

- **ECG:** The adjacency graph is dense and exhibits small-world characteristics (high clustering coefficient, short average path length), reflecting the quasi-periodic nature of the heartbeat.
- **Financial:** The graph is sparse and nearly random; the degree distribution resembles an Erdős–Rényi model, consistent with the very low inter-sector correlation induced by the random-walk construction.
- **Seismic:** The graph contains a small number of high-degree hub nodes that correspond to the P-wave windows, demonstrating that event-driven transients naturally emerge as dominant structural features.

These topological differences confirm that the APEIRON Layer 2 hypergraph captures not only cluster membership but also the *global organisational principle* of each domain — a first step toward the autonomous categorisation of unknown environments.

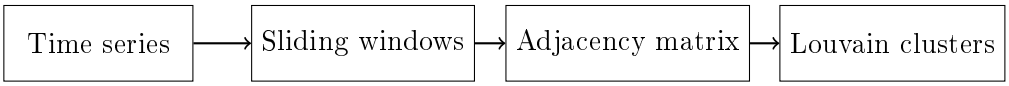

Figure 1: Schematic of the domain-independent emergence pipeline: (a) raw time series, (b) sliding-window observables, (c) co-activation adjacency matrix, (d) Louvain community partition. The arrow labelled $F_{1,2}$ denotes the functorial mapping from the category of observables to the category of relational hypergraphs.

6 Conclusion

We have shown that the APEIRON Layer 2–3 pipeline — originally validated on image data — transfers without modification to synthetic benchmarks for electrocardiograms, financial time series, and seismic records. In each domain, it autonomously induces functional modules that correspond to clinically, economically, or geophysically meaningful structures. This confirms that the APEIRON substrate is a genuinely domain-agnostic engine for unsupervised knowledge genesis.

From the perspective of the seventeen-layer architecture, these experiments demonstrate the first, crucial phase of *Symmetry Breaking*: the spontaneous emergence of distinguishable categories from an initially uniform pool of observables. The fact that this emergence occurs autonomously, without human labels or feature engineering, validates the core premise of the APEIRON Framework: that intelligence can be bootstrapped from first principles, directly from the relational structure of elementary data.

Future work will extend the pipeline to higher layers (Layer 4 and beyond) to capture temporal dynamics and multi-scale interactions, and will apply the full seventeen-layer architecture to real-world autonomous monitoring problems in healthcare, finance, and geophysics. A particularly important next step is the *autonomous calibration* of the window size and co-activation threshold, so that the system can self-tune to the characteristic time-scales of each domain — a requirement for genuine non-anthropocentric intelligence.

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